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Roll No

FT-5003-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Fire Prevention and Protection Measures

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What are the fire protection systems must be installed in assembly building with sub classification D4 as per national building code?
b) Explain the standard temperature time test of fire exposure for the structural components of building?
2. a) Give a detail description on types of building construction with NFPA sub type of building?
b) What are the general requirements for evacuation facilities within a multistory building? Explain the importance of fire tower over the stair case?
3. a) Explain the classification of fire alarm system based on working principle.
b) Explain phased evacuation of the building.
4. a) Explain flaming combustion and extinguishment by its own combustion products of a wax candle with suitable diagram.

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- b) Compare the fire extinguishment process with water and water mist on common fire by considering suitable fuel.
- 5. a) Explain the different types and subtypes of automatic fire detectors used in different occupancy.
b) Write the visual inspection schedule for fire alarm system as per NFPA-72.
- 6. a) Explain the working principle of gas and explosive mixture detector with suitable diagram.
b) Explain working principle of ionization smoke detector with neat and clean sketch.
- 7. a) Explain fire protection and life safety requirement underground structure.
b) Explain life safety requirement for building under construction.
- 8. a) Define refuse area and assembly points. Discuss the role and importance of both in evacuation procedure.
b) Explain fire extinguishment with dry chemical powder based on composition, particle size and chain breaking mechanism.
